

MTH 240 — Linear Algebra

Department of Mathematics · Fall 2026 · Sections 01–04

Course Coordinator

Dr. Yuki Tanaka, ytanaka@example.edu, VIC 606. Section instructors are listed on the course portal.

Course Materials

There is no cost associated with the materials for this course. The textbook, *Linear Algebra Done Openly*, is an open educational resource released under a Creative Commons licence and is available as a free PDF from the course portal. A printed copy may be ordered at cost from the campus print shop if you prefer paper, but this is entirely optional and no student is expected to do so.

Software: computations use Octave and Python, both of which are free and are installed on all lab machines. No licence purchase is required. No clicker or online homework subscription is used in this course.

Assessment

Component	Weight	Date
Online homework (11 sets)	10%	Weekly, Sundays 23:59
Term Test 1	20%	Thursday, October 8, 2026
Term Test 2	20%	Thursday, November 12, 2026
Final Examination	50%	Friday, December 11, 2026

Topics by Week

Week	Topic
1 (Sept 8)	Systems of linear equations; Gaussian elimination
2 (Sept 14)	Matrix algebra; inverses
3 (Sept 21)	Determinants
4 (Sept 28)	Vector spaces and subspaces
5 (Oct 5)	Basis and dimension
Reading week	October 12–16 — no classes
6 (Oct 19)	Linear transformations
7 (Oct 26)	Rank–nullity theorem
8 (Nov 2)	Eigenvalues and eigenvectors
9 (Nov 9)	Diagonalization
10 (Nov 16)	Inner product spaces
11 (Nov 23)	Orthogonality; Gram–Schmidt
12 (Nov 30)	Least squares; applications

Week	Topic
13 (Dec 7)	Review

Policies

Term tests missed for documented medical reasons have their weight transferred to the final examination. There are no make-up tests. Calculators are not permitted on term tests or the final examination; you will not need one.

Important dates: classes begin September 8; last day to drop with full refund September 21; 50% refund until October 2; last day to drop without academic penalty November 10; last day of classes December 7; examination period December 9–18.